

A Generalizable Semantic Graph Learning Framework for Road Network Representation

Yonghui Liu

Cho Chun Shik Graduate School of Mobility
Korea Advanced Institute of Science and Technology
Daejeon, South Korea
liuil@kaist.ac.kr

Inhi Kim

Cho Chun Shik Graduate School of Mobility
Korea Advanced Institute of Science and Technology
Daejeon, South Korea
inhi.kim@kaist.ac.kr

Abstract—Road networks (RNs) are fundamental to transportation systems, serving as critical input for applications like traffic prediction and autonomous vehicle navigation. However, effectively representing large-scale RNs is challenging due to their complex topology, dynamic evolution, and heterogeneous associated data. Existing approaches often struggle with scalability, generalization to new cities, and integrating contextual semantic information. In this work, we propose HetSemNet, a heterogeneous graph framework that enriches road network graphs with semantic context nodes derived from point-of-interest (POI) descriptions encoded by a pre-trained large language model (LLM). On this graph, we introduce SaRNSAGE, a semantic-aware graph neural network that employs a sampling-and-aggregation mechanism to learn robust and transferable road segment representations. SaRNSAGE jointly captures road connectivity and surrounding semantic context, leveraging basic road attributes as initial features while avoiding complex feature engineering. Experiments on real-world urban road networks demonstrate that our method significantly outperforms conventional GNN baselines on a road type classification task and exhibits strong generalization in a cross-city transfer scenario. These results underscore the effectiveness of combining LLM-derived semantic information with heterogeneous graph modeling for generalizable road network representation learning.

Index Terms—Transportation Networks, Data Fusion, Heterogeneous Graphs, Large Language Models, Graph Neural Networks, Representation Learning

I. INTRODUCTION

Road networks are essential infrastructure for transportation, supporting daily activities such as commuting, logistics, and urban mobility. Many intelligent transportation applications, like traffic forecasting [8] and autonomous vehicle decision-making [7], [13], depend on effective road network representations. However, representing large-scale road networks, which are complex, dynamic, and encompass heterogeneous data (e.g., geometric attributes, traffic data, and surrounding points of interest), is a non-trivial challenge.

Traditional approaches rely on handcrafted features (e.g., road length, speed limit) to represent road segments. While effective in specific scenarios, these static features lack adaptability and generalizability, especially when road networks change or when models are applied to new cities [3], [14].

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2021R1C1C1006405). (Corresponding author: Inhi Kim).

These feature-engineered methods often fail to capture the dynamic and contextual aspects of road usage.

Graph-based learning methods, such as Graph Convolutional Networks (GCNs) and GraphSAGE, offer an alternative by modeling road networks as graphs. In this representation, nodes represent road segments (or intersections), and edges represent connectivity. GNNs can learn embeddings for each road by aggregating information from neighboring nodes, capturing complex structural patterns [5]. Despite their success in tasks like traffic speed prediction and route optimization, GNNs face limitations in scalability and the lack of semantic context. Existing models often ignore rich contextual data, such as nearby businesses or amenities, which could provide valuable insights into road function and usage.

Recent advancements in Large Language Models (LLMs) like BERT and GPT have enabled the encoding of textual data into semantic embeddings that capture nuanced context. Studies have shown that LLM-based text analysis can enhance transportation modeling by revealing semantic information about road surroundings. However, integrating this unstructured semantic data with road network models remains challenging, as concatenating text-derived features can introduce high dimensionality and noise.

To address these challenges, we propose a novel framework, HetSemNet + SaRNSAGE, for semantic-aware road network representation learning. The key idea is to construct a heterogeneous graph that integrates both road segments and contextual semantic nodes, learning road representations using a specialized GNN that fuses topological and semantic information. The main contributions of this paper are:

- **Semantic-enriched Heterogeneous Graph (HetSemNet):** We introduce a graph representation that incorporates semantic context by adding POI-based context nodes connected to road nodes based on spatial proximity. Each context node is annotated with a text description, encoded into a vector using a pre-trained LLM, enriching the graph with information about nearby places.
- **Semantic-Aware GNN (SaRNSAGE):** We propose a new GNN architecture, extending GraphSAGE to handle multiple node types and feature sources. SaRNSAGE uses a dual-aggregation approach: aggregating information from road neighbors and context neighbors separately.

It also concatenates raw context embeddings with aggregated features, preserving critical semantic information.

- **Generalization and Transferability:** Our inductive framework can generate embeddings for new roads or cities without retraining. We demonstrate that a model trained on one city’s road network can be applied to another city, achieving high performance without fine-tuning, making it suitable for real-world deployment where extensive labeled data is impractical.

We evaluate the approach on a road type classification task using real urban road network data. The results show that HetSemNet + SaRNSAGE outperforms traditional methods and state-of-the-art GNN baselines. The model also maintains its advantage in cross-city evaluations, confirming its improved generalizability. This work is one of the first to integrate LLM-derived semantic embeddings into a heterogeneous GNN for road networks and demonstrates strong transfer learning performance across cities.

II. RELATED WORK

A. Road Network Representation Learning

Fig 1 illustrates a schematic road network that can be transformed into different graph representations, including intersection-centric and road-centric network graphs [2], [16]–[18]. Early studies on network embedding, such as DeepWalk [9], LINE [10], and node2vec [4], were applied to road networks to capture structural features in low-dimensional vectors. These methods perform unsupervised learning via random walks or matrix factorization to encode graph structure but typically do not utilize node attributes or external data.

More recent approaches leverage graph neural networks (GNNs) for road networks. For example, Graph Convolutional Networks (GCNs) [6] and GraphSAGE [5] can learn road segment representations by aggregating information from connected roads, capturing the local topology. Variants like Graph Attention Networks (GATs) [11] introduce attention mechanisms to weight important neighbors. While GNNs improve over static embeddings by incorporating connectivity and attributes, applying them to large-scale, evolving road networks is challenging. Existing GNN models often assume a fixed graph; if the road network changes, the model must be retrained. Moreover, most prior road network GNN studies focus on purely structural information or simple attributes (e.g., traffic volume) and do not integrate rich semantic context from the environment.

B. Incorporating Semantics and Context

To enhance road representations, recent works have explored including additional data. Some methods incorporate traffic patterns or trajectories associated with roads to inform the embeddings. Others consider geographic context: for instance, Chen et al. use “traveling semantics” and traffic patterns to learn robust road embeddings [2], and Zhang and Long propose a dual-graph approach separating road connectivity and functional similarity [18]. These approaches indicate the benefit of context but typically rely on structured data

(traffic counts, trajectories) rather than unstructured semantic information.

The use of textual data in transportation has been limited mostly to analysis tasks (like sentiment analysis of transport-related tweets [1] or parsing accident reports [15]) rather than embedding it directly into network models. Large Language Models (LLMs) offer a powerful way to capture semantics from text descriptions (such as the names of nearby POIs or neighborhoods). However, fusing LLM-based text embeddings with graph neural networks is non-trivial; it requires handling high-dimensional features and potential noise. Our work is closely related to the emerging idea of heterogeneous graph learning for transportation. We adopt a heterogeneous graph approach similar in spirit to Heterogeneous Graph Attention Networks (HAN) [12], but tailored for road networks and augmented with LLM-derived node features. In contrast to prior methods, our framework explicitly leverages unstructured textual context around roads and demonstrates that this leads to more generalizable representations.

C. Summary

The literature review highlights substantial advancements and key limitations in road network representation learning and textual embeddings facilitated by large language models. Traditional embedding methods and GNN frameworks have significantly improved network analysis but continue to face challenges related to heterogeneity, scalability, and semantic integration. Concurrently, LLM-driven textual embeddings offer powerful semantic representations but necessitate sophisticated methods to effectively integrate and manage their complexity within network frameworks. Addressing these gaps motivates our proposed framework, aiming to enhance generalizability, semantic richness, and computational efficiency in road network embeddings.

III. PROPOSED METHODOLOGY

A. Overview

The proposed SaRNSAGE framework (illustrated in Fig. 2) comprises two main components: heterogeneous graph construction and a semantic-aware graph aggregation model. The raw road network is first transformed into a heterogeneous graph, termed *HetSemNet*, where road segments and contextual entities (e.g., POIs) are represented as distinct node types. Road nodes are initialized with structural attributes, while context nodes are assigned semantic embeddings extracted from textual descriptions using a pretrained large language model (LLM). This unified graph representation captures both topological structure and semantic context.

Based on this graph, SaRNSAGE learns node embeddings via neighborhood sampling and dual-path aggregation. Road and context neighbors are treated separately to avoid over-smoothing and semantic dilution. To preserve semantic integrity, the model further injects the raw average of context embeddings as an auxiliary signal. This architecture is designed to support both supervised and unsupervised training,

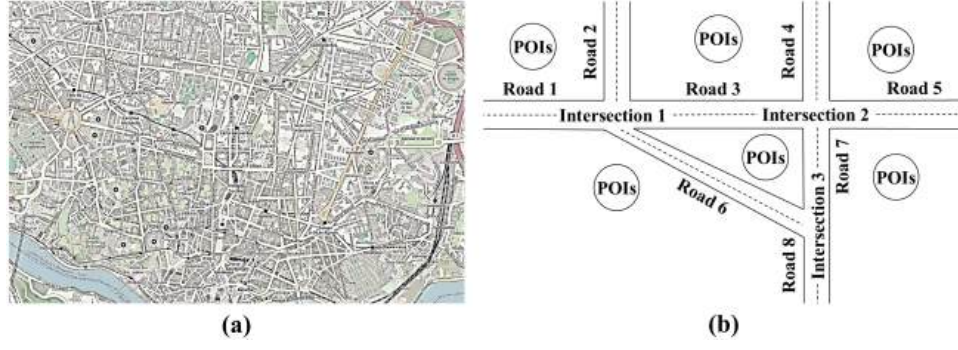


Fig. 1. A sketch of road network. (a) Real road network visualization from OSM. (b) Schematic representation of roads, POIs, and intersections.

enabling effective adaptation to diverse urban settings with varying levels of data availability.

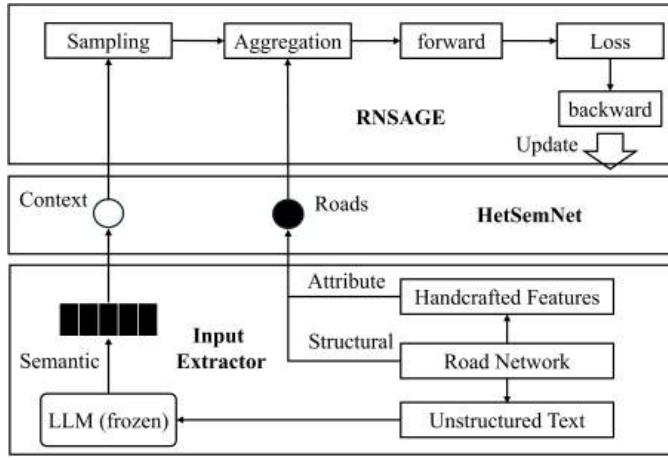


Fig. 2. Framework of the proposed SaRNSAGE.

B. HetSemNet: Heterogeneous Graph Construction

To model the rich structural and contextual dependencies of road networks, we construct a heterogeneous graph defined as:

$$G = (V, E, A_v, A_e, R)$$

where V is the node set, E the edge set, A_v and A_e denote node and edge attributes, and R specifies the edge types. The node types include:

- **Road Nodes** (v_r): Represent individual road segments, each associated with structural attributes such as node degree, segment length, and speed limit.
- **Context Nodes** (v_c): Represent nearby facilities or POIs, enriched with semantic information derived from text metadata.

Edges are categorized as:

- (Road, connects, Road): Encodes direct topological connectivity between adjacent road segments.
- (Road, connects, Context): Links a road segment to nearby context nodes within a spatial search radius D .

Context nodes are not connected to each other, preserving a localized semantic influence. Each road-context edge is weighted using a distance-sensitive exponential decay function:

$$w = \exp\left(-\frac{\text{dis}(v_r, v_c)^2}{\gamma^2}\right) \quad (1)$$

where $\text{dis}(v_r, v_c)$ denotes the spatial distance and γ controls the decay rate, with larger values allowing distant nodes to contribute more.

C. Pre-trained Semantic Embedding

Each context node is paired with a descriptive text s_i , such as a facility name or category. This text is encoded using a pretrained language model to produce a fixed-dimensional semantic embedding:

$$c_i = \text{LLM}(s_i), \quad c_i \in \mathbb{R}^{d_c} \quad (2)$$

These embeddings remain static throughout training to preserve the general semantic structure learned from large-scale corpora. In cases where descriptive text is missing, we substitute with placeholder tokens or use category-level average embeddings.

Figure 3 illustrates the semantic extraction pipeline that generates LLM-based embeddings for POI nodes in the heterogeneous graph.

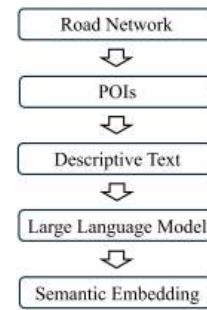


Fig. 3. Workflow of semantic embedding extraction from the road network. This module encodes POI metadata into high-dimensional embeddings using a pretrained LLM.

D. SaRNSAGE: Sampling and Aggregation on HetSemNet

Given the HetSemNet graph and input features, SaRNSAGE computes embeddings for road nodes through multi-hop aggregation. To balance representation fidelity and computational cost, all connected road neighbors are retained, while a fixed number K of context neighbors are sampled for each node. Let $N_{v,r}$ denote the set of road neighbors and $N_{v,c}^K$ the sampled set of context neighbors. At layer k , the aggregation is defined as:

$$m_{v,r}^k = \text{AGG}^r(\{h_u^{k-1} \mid u \in N_{v,r}\}) \quad (3)$$

$$m_{v,c}^k = \text{AGG}^c(\{h_u^{k-1} \mid u \in N_{v,c}^K\}) \quad (4)$$

$$m_{\text{avg},c}^k = \text{AVG}^c(\{c_w \mid w \in N_{v,c}^K\}) \quad (5)$$

These components are concatenated to form the input for the node update:

$$m_v^k = [m_{v,r}^k \parallel m_{v,c}^k \parallel m_{\text{avg},c}^k] \quad (6)$$

$$h_v^k = \sigma(W^k m_v^k + b^k) \quad (7)$$

Here, W^k, b^k are learnable parameters, and σ is a non-linear activation function (e.g., ReLU). The embedding of context nodes remains unchanged across layers:

$$h_{v,c}^k = c_i \quad (8)$$

This dual-path aggregation scheme enables the model to incorporate both learned and raw semantic signals. Empirically, we observe that injecting unprocessed semantic averages improves the robustness and interpretability of the resulting road representations.

Figure 4 illustrates the aggregation process in SaRNSAGE. Each road node aggregates features from its structural neighbors (black) and a sampled subset of nearby semantic context nodes (gray), effectively reducing noise and maintaining scalability.

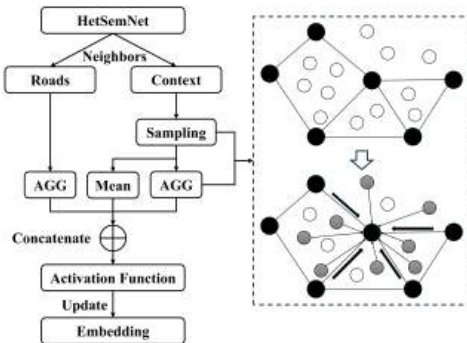


Fig. 4. Road network sampling and aggregation. SaRNSAGE aggregates information from structural neighbors (black) and sampled semantic context nodes (gray). The sampling mechanism avoids full context propagation (white), enhancing scalability.

IV. EXPERIMENTS AND DISCUSSION

We evaluate the proposed SaRNSAGE model on a road type classification task using real-world road network data. The experiments are designed to assess both within-city performance and cross-city generalizability, as well as the model's scalability and computational efficiency.

A. Model Settings

Road segments were partitioned into training, validation, and testing sets using a 7:2:1 ratio. Table I summarizes the hyperparameter settings for HetSemNet construction, the SaRNSAGE architecture, and the MLP classifier used for downstream prediction.

We compare SaRNSAGE with the following baseline and state-of-the-art models:

- **MLP with Handcrafted Features:** Uses basic attributes (e.g., node degree, road length) as direct input to a shallow MLP classifier.
- **GCN (Graph Convolutional Network):** Aggregates node features over a homogeneous road-only graph.
- **GAT (Graph Attention Network):** Enhances GCN by learning attention weights over neighbors.
- **GraphSAGE:** Uses sampling-based aggregation for improved scalability.
- **SaMEAN:** Aggregates semantic embeddings from connected POIs via a simple mean operator.
- **SaHAN:** A heterogeneous attention network that learns node-type-aware interactions across HetSemNet.

It is important to note that GCN, GAT, and GraphSAGE operate on homogeneous graphs with only road nodes, whereas SaHAN and SaRNSAGE are designed for heterogeneous graphs and fully leverage both topological structure and contextual semantics provided by POI nodes.

TABLE I
MODEL PARAMETER SETTINGS

Module	Parameter Name	Value
HetSemNet	LLM	bert-base-multilingual-cased
	d_c	768
	γ	640
SaRNSAGE	K	128
	k	1
	AGG^r	Mean Pool
	AGG^c	Mean Pool
	d	128
	Batch Size	512
MLP	Learning Rate	0.001
	Hidden Layers	(64,32)
	Activation	ReLU

B. Model Training and Evaluation

We adopt a contrastive learning strategy for unsupervised road representation learning. Training data are generated based

on the structure of unlabeled road networks. For each road node $v_{r,i}$, we randomly sample one positive neighbor $P_1 \in N(v_{r,i})$ and five negative nodes $\{N_1, \dots, N_5\} \notin N(v_{r,i})$. This forms a set of positive training pairs $\{(v_{r,i}, P_1)\}$ labeled as 1 and negative pairs $\{(v_{r,i}, N_j)\}_{j=1}^5$ labeled as 0.

The model is trained using binary cross-entropy (BCE) loss:

$$\mathcal{L}_{\text{BCE}} = -(y \log p + (1 - y) \log(1 - p)) \quad (9)$$

where $y \in \{0, 1\}$ is the ground truth label and $p \in [0, 1]$ is the predicted probability of the node pair being similar.

Once the SaRNSAGE encoder is trained, the learned node embeddings are frozen and used to train a multi-layer perceptron (MLP) classifier for road type prediction. This classification phase is used strictly for evaluation. Model performance is evaluated using micro-averaged Precision, Recall, and F1-score over all classes.

C. Performance Comparison

We first assess classification performance within a single-city setting (Porto). Table II reports the results in terms of Precision, Recall, and F1-score. As expected, all graph-based models surpass the MLP baseline (F1 = 0.4835), confirming the importance of topological context in learning road representations. Among them, GraphSAGE achieves a 6.78% improvement and GAT a 7.80% improvement in F1-score.

When semantic context is added, performance improves substantially. SaMEAN achieves a 29.53% boost over the MLP baseline, demonstrating that even simple semantic pooling can enhance prediction accuracy. SaHAN further improves upon SaMEAN by 19.32%, benefiting from its attention mechanism over heterogeneous neighbors.

Our proposed SaRNSAGE model yields the best overall results, achieving an F1-score of 0.7296. This represents a 50.90% improvement over the MLP baseline and a 16.50% improvement over SaMEAN. These results confirm that modeling both structural and semantic contexts—especially with semantic preservation and dual-path aggregation—leads to more expressive and robust road representations.

Furthermore, the consistent gains across both homogeneous and heterogeneous settings suggest that SaRNSAGE effectively balances the benefit of semantic enrichment without compromising topological information flow.

TABLE II
CLASSIFICATION ACCURACY COMPARISON ACROSS METHODS

Methods		Precision	Recall	F1
w/o Semantic	Handcrafted	0.4359	0.6155	0.4835
	GCN	0.4517	0.6191	0.5001
	GraphSAGE	0.5110	0.6221	0.5163
	GAT	0.5053	0.6257	0.5212
Semantic-aware	SaMean	0.6507	0.6792	0.6263
	GCN	0.6929	0.7154	0.6860
	GraphSAGE	0.6939	0.7164	0.6878
	GAT	0.6983	0.7201	0.6939
	HAN	0.6675	0.6991	0.6577
	SaRNSAGE	0.7282	0.7458	0.7296

D. Running Efficiency

To understand the trade-off between classification performance and computational cost, we evaluate the impact of context sampling size K . Figure 5 shows the F1-score and training time for different values of K .

Results show that increasing K improves F1-score initially but with diminishing returns. The peak performance is reached at $K = 2^9$, beyond which gains are minimal. Meanwhile, training time increases rapidly as K grows, due to the additional context nodes processed during aggregation.

This experiment highlights the practical utility of context sampling in SaRNSAGE. With a moderate sample size (e.g., $K = 2^6$), the model achieves competitive accuracy while keeping computational demands acceptable. The setting marked “All,” where all POIs are used without sampling, does not significantly improve accuracy but results in a much higher cost. This underscores the design rationale of SaRNSAGE: leveraging POI semantics efficiently via stochastic sampling.

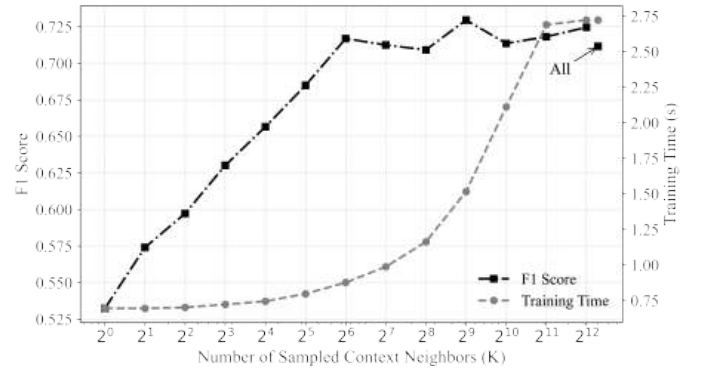


Fig. 5. F1-score and training time as a function of sampled context neighbors K . Sampling improves scalability while maintaining accuracy.

E. Adaptivity and Generalizability

SaRNSAGE is not only accurate but also designed for deployment in dynamic urban environments. Its sampling-based architecture enables local embedding updates when new roads or POIs are added, avoiding costly retraining on the entire graph. This makes it well-suited for scalable, real-time traffic and infrastructure applications.

To evaluate generalization capability, we conduct a cross-city transfer experiment: the model is trained on Porto and tested on Lisbon. Figure 6 illustrates the two cities, which differ significantly in layout, size, and facility distributions. Table III shows that the model achieves an F1-score of 0.6766 in the transfer setting (Porto→Lisbon), only 4.71% lower than its in-domain performance (Porto→Porto, F1 = 0.7296). When compared to a model trained directly on Lisbon (F1 = 0.6968), the transferred model incurs only a 2.90% performance loss.

These findings validate the robustness and transferability of SaRNSAGE. Despite urban disparities, the model retains its effectiveness, benefiting from both semantic generalization (via static pretrained embeddings) and structural adaptability

(via local sampling). This is particularly valuable in low-resource cities where annotated data are scarce or in fast-evolving environments requiring continual updates.

TABLE III
RESULTS FOR ROAD TYPE CLASSIFICATION ACROSS CITIES

Training Area	Test Area	Precision	Recall	F1
Porto	Porto	0.7282	0.7458	0.7296
Lisbon	Lisbon	0.7024	0.7143	0.6968
Porto	Lisbon	0.6765	0.6936	0.6766

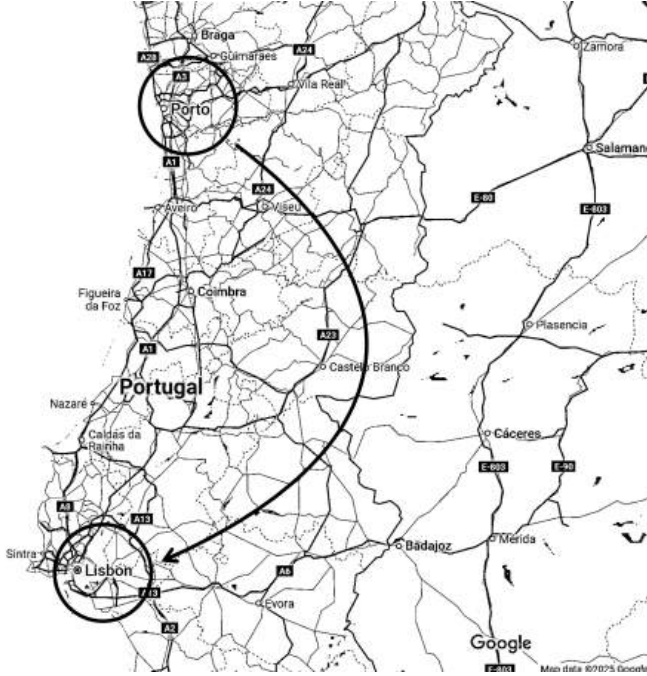


Fig. 6. Geographical locations of Porto and Lisbon in Portugal, used as distinct urban testbeds for cross-city evaluation.

V. CONCLUSION

This paper presents SaRNSAGE, a semantic-aware graph representation learning framework designed to enhance road network modeling by integrating structural topology and unstructured contextual semantics. Built upon a heterogeneous graph structure (HetSemNet), the framework encodes road segments and surrounding points of interest (POIs) as distinct node types, with semantic features derived from pretrained large language models.

Through a dual-path aggregation mechanism, SaRNSAGE effectively fuses both structural and semantic signals, while a context sampling strategy ensures scalability to large and dense urban networks. Experimental results on a multi-class road type classification task demonstrate that SaRNSAGE significantly outperforms both traditional feature-based methods and state-of-the-art graph baselines. Importantly, the model exhibits strong generalization ability, achieving robust performance when transferred to unseen cities without retraining.

REFERENCES

- [1] Farman Ali, Shaker El-Sappagh, and Daehan Kwak. Transportation sentiment analysis using word embedding and ontology-based topic modeling. *Knowledge-Based Systems*, 174:27–42, 2019.
- [2] Yile Chen, Xiucheng Li, Gao Cong, Zhifeng Bao, Cheng Long, Yiding Liu, Arun Kumar Chandran, and Richard Ellison. Robust road network representation learning: When traffic patterns meet traveling semantics. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*, pages 211–220. ACM, 2021.
- [3] Mihaela Condurat, Alexandru M Nicuță, and Radu Andrei. Environmental impact of road transport traffic: a case study for county of iași road network. *Procedia Engineering*, 181:123–130, 2017.
- [4] Aditya Grover and Jure Leskovec. node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 855–864. ACM, 2016.
- [5] William L Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, volume 30, pages 1024–1034, 2017.
- [6] Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2017.
- [7] Shen Li, Keqi Shu, Chaoyi Chen, and Dongpu Cao. Planning and decision-making for connected autonomous vehicles at road intersections: A review. *Chinese Journal of Mechanical Engineering*, 34(1):133, 2021.
- [8] Xiaolei Ma, Haoran Zhong, Yuxuan Li, Jian Ma, Zhen Cui, and Yinhai Wang. Forecasting transportation network speed using deep capsule networks with nested lstm models. *IEEE Transactions on Intelligent Transportation Systems*, 22(8):4813–4824, 2021.
- [9] Bryan Perozzi, Rami Al-Rfou, and Steven Skiena. Deepwalk: Online learning of social representations. In *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 701–710. ACM, 2014.
- [10] Jian Tang, Meng Qu, Mingzhe Wang, Ming Zhang, Jun Yan, and Qiaozhu Mei. Line: Large-scale information network embedding. In *Proceedings of the 24th International Conference on World Wide Web*, pages 1067–1077. International World Wide Web Conferences Steering Committee, 2015.
- [11] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, and Yoshua Bengio. Graph attention networks. *arXiv preprint arXiv:1710.10903*, 2018.
- [12] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, and P.S. Yu. Heterogeneous graph attention network. In *Proceedings of the World Wide Web Conference*, pages 2022–2032. ACM, 2019.
- [13] Yifan Wang, Peng Cai, and Guo Lu. Cooperative autonomous traffic organization method for connected automated vehicles in multi-intersection road networks. *Transportation Research Part C: Emerging Technologies*, 111:458–476, 2020.
- [14] Chao-Yun Wu, Mao-Bin Hu, Rui Jiang, and Qing-Yi Hao. Effects of road network structure on the performance of urban traffic systems. *Physica A: Statistical Mechanics and its Applications*, 563:125361, 2021.
- [15] Nan Xu, Lei Ma, Qian Liu, Lei Wang, and Yong Deng. An improved text mining approach to extract safety risk factors from construction accident reports. *Safety Science*, 138:105216, 2021.
- [16] Cheng Yang, Zhiyuan Liu, Deli Zhao, Maosong Sun, and Edward Y Chang. Network representation learning with rich text information. In *Proceedings of the 24th International Joint Conference on Artificial Intelligence*, pages 2111–2117, 2015.
- [17] Daokun Zhang, Jian Yin, Xingquan Zhu, and Chengqi Zhang. Network representation learning: A survey. *IEEE Transactions on Big Data*, 6(1):3–28, 2020.
- [18] Liang Zhang and Cheng Long. Road network representation learning: A dual graph-based approach. *ACM Transactions on Knowledge Discovery from Data*, 17(9):121:1–121:25, 2023.